Discussion

What the Template Is, and Is Not I

The pipeline measures the *shape* of a literature — its size, growth, subfield composition, topical structure, citation geometry, and the hypotheses a field frames. It does not adjudicate scientific truth. The optional hypothesis scores summarize how the retrieved corpus *talks about* each claim, weighted by citation influence; they are an evidence-landscape instrument, not a verdict.

What the Template Is, and Is Not I

The 5 topics extracted by NMF provide a data-driven complement to the keyword-based subfield taxonomy. Where the taxonomy assigns each paper to a single bucket, the topics reveal overlapping thematic structure: a paper on modafinil's cognitive effects in ADHD patients belongs to the "Psychiatry" subfield but also loads on the "Cognitive Enhancement" and "ADHD Treatment" topics. This multi-resolution view is more informative than either approach alone.

The committed analysis corpus is a bounded retrieval snapshot. It predates the deterministic per-engine retrieval report introduced by this template, so this paper does not attribute record counts or success/failure states to individual engines by reverse-engineering the merged corpus. New runs write those facts directly to `output/data/retrieval_report.json`; a resumed legacy snapshot is explicitly labelled `resume_without_prior_retrieval_report` rather than being given invented provenance.

Engine Coverage and Bias I

The max-results cap of 1,000 per engine means the full literature is larger than the retrieved corpus; the 2334 records represent a bounded sample rather than the complete literature. The citation network resolution rate of 22.4% reflects this: many cited works lie outside the retrieved slice. Increasing the cap or adding more engines would improve coverage but also increase runtime and API load.

Honest Defaults I

The small corpus under `data/fixtures/` is synthetic (reserved test DOIs and generated authors) and exists only for offline tests. It is not silently substituted for the tracked analysis corpus. A user who regenerates empirical findings must run retrieval, analysis, figures, and manuscript injection together and retain the resulting corpus and retrieval report; fixture-only runs demonstrate machinery, not findings about modafinil.

Limitations and Extensions I

Several limitations bound the interpretation of results:

Limitations and Extensions I

- **Coverage is bounded by the enabled engines and the query.** The max-results cap truncates each engine's contribution. Semantic Scholar's rate limiting excluded a major source; a Semantic Scholar API key would resolve this.
- **Subfield classification is keyword-based** and only as good as the configured taxonomy. Ambiguous papers may be misclassified; a classifier based on embeddings or supervised learning could improve accuracy.
- **The default embeddings are lexical** (TF-IDF/SVD). They capture term co-occurrence but not semantic similarity; a transformer backend (`embeddings` extra) would improve the quality of nearest-neighbour retrieval and clustering.
- **Hypothesis scoring depends on an external language model.** Without Ollama running, scores read *pending*; with Ollama configured (as in this run, with 127 assertions extracted), scores are populated. The scoring is also sensitive to prompt design and model choice; the default `gemma3:4b` is a lightweight model suitable for demonstration but may miss nuanced assertions.
- **Abstract coverage is 61.6%.** Text analytics operate only on the subset of records with abstracts, biasing topic models and embeddings toward well-indexed sources.

Limitations and Extensions I

Each limitation is a configuration or dependency choice rather than a change to the core architecture.